Printavo Hub — Architecture & Systems Overview

Mint Printworks · August 29, 2026 · Prepared for external technical review · Companion to the Project Plan & Status sheet

What this is

Printavo Hub is one consolidated internal platform for Mint Printworks (screen printing / embroidery shop) replacing 11 separate Printavo-connected apps. Printavo is the shop's system of record (quotes, invoices, customers, payments, production statuses); the Hub continuously mirrors it into its own Postgres database and builds every module on that local mirror — reporting, purchasing, production printing, quoting, client intelligence, email correspondence, and AI assistance. Core design stance: **Printavo's API is slow and rate-limited, so read paths never touch it live; write paths are few, guarded, and human-confirmed.**

Platform Architecture

- **Monorepo** (pnpm workspaces, Node 24, TypeScript): **Hub** — React 18 + Vite SPA (staff-facing); **API server** — Express 5, all business logic and background schedulers; **moMINTum** — ported task-app frontend served alongside; shared libs for DB schema, Printavo client, API spec.
- **Data**: PostgreSQL + Drizzle ORM. ~40 schema modules: pv_* Printavo mirror tables, sync bookkeeping (runs, checkpoints, watermarks, change events), plus Hub-owned domains (purchasing, print queue, catalog, branding, correspondence, AI drafts…). Hub-owned data is **never** written into mirror tables — enrichment lives in separate tables keyed by Printavo ID.
- **API contract**: OpenAPI spec is the source of truth; Orval codegen produces typed React Query hooks + Zod schemas consumed by the frontend. No hand-written fetch calls.
- **Auth**: Clerk (Google/email sign-in) + server-side staff allowlist. Roles: admin vs staff, plus per-capability grants (e.g. email activity). Sensitive actions add an admin PIN. Every API route sits behind staff auth; email/correspondence adds a per-mailbox ACL layer.
- **Hosting**: Replit (dev + published production as separate environments, separate databases). Files (mirrored Printavo attachments, PNGs, email images) live in object storage.

Foundation — Printavo Sync Engine

- Printavo GraphQL v2 has **no updated-after filter**, so incremental sync re-crawls invoices+quotes every 5 min (rolling 12-month window) and detects changes by **content hash** after timestamp normalization (Printavo renders the same instant in varying timezones). ~2.3 min per poll, ~23% of the account-wide 10 req/5s budget.
- Detail backfill (line items, imprints), payments history (2017→today), messages, inquiries, merch stores, reference data, and order files mirrored to object storage. Full-history backup to Feb 2018 (~19k orders) with one-click portable ZIP export.
- Self-healing scheduler: interrupted runs resume at boot; deep re-checks nightly; global sync bar + manual "Sync Now" on every page.
- Change events (created / updated / status_change / converted) feed downstream automation like status rules and task sync.

Functional Modules (current)

Analyze

Reports API replicating legacy Insights formulas (invoice-only revenue, cash-basis payments) · Orders & Payments explorers · Pipeline and Status Activity (arrivals, dwell, occupancy) · **Ask AI**: plain-English questions → guarded LLM-written SQL (details under Safety), answer shown with table + SQL, saveable as deterministic presets.

Clients (Client 360)

~3,565-customer directory ranked by spend, niche categories + niche report, customer profiles (lifetime/YTD spend, frequency, top styles, decoration mix, contacts, order history) · **Brand standards**: brand colors (Pantone/hex), logo assets, artwork bookmarks pulled read-only from the shop's Dropbox with invoice-number links back to Printavo · Garment-color collections with Pantone catalog + palette suggestions.

Purchasing / Vendor Orders

Needs-ordering scan of open orders → cart → PO builder with live SanMar + S&S pricing and warehouse auto-pick → push line-item updates + PO PDF back to Printavo → **electronic order submission to both vendors (live-proven)** with PIN gate, kill switch, and anti-double-order guards → tracking dashboard. Status-rules engine suggests order-status moves; staff confirm each.

Production — Print Queue

Status-filtered workorder printing with presets and combined previews · bulk status changes with stale-order guards · auto-print to physical printers via PrintNode (dry-run default, double-print protection) · in-house packing slips / box labels rendered from local data (near-clone of Printavo's template).

Quoting

Price Calculator (faithful port of the legacy Breezy Quote tool: decoration matrices, exact legacy math, PNG export) · **AI Quote Drafts**: paste a customer request, AI extracts a structured draft quote priced from live matrices; staff review/edit; creating the real Printavo quote requires an explicit typed confirmation, idempotency-keyed, one attempt only.

Email & Correspondence

Read-only Gmail monitoring of CSR mailboxes (service-account, metadata-only) mapped to customers/invoices; reply-tracking instead of read receipts; email-effort deals report · Missive correspondence sync into a per-customer communication timeline · AI search over correspondence with citation validation — **ACL is applied before any AI retrieval**, plus an AI kill switch.

Shared-DB sister apps

moMINTum (task manager) and **Mint Bucks** (store credit) run inside the Hub against the standalone apps' **live production databases** — email-only identity mapping, additive-only hub-owned tables, advisory-lock JIT creation, write-blocked shared settings, and phased ownership of pollers (dormant until flipped). Two-way Printavo ⇄ moMINTum task sync with twin routing and completion reconciliation.

Also

Chipply web-store imports (revenue kept strictly separate from Printavo revenue; invoices link themselves via nickname convention) · Shipping rate estimator (ShipStation + EasyPost side-by-side) and packing-slip/label sorter · Product Finder (cross-vendor stock/catalog search) · Backup dashboard.

Integrations

SERVICE	USED FOR	ACCESS MODEL
Printavo (GraphQL v2)	System of record — sync, write-backs, quotes, tasks	API token · throttled shared gateway (account-wide 10 req/5s)
SanMar	Live pricing/inventory, catalog import, electronic POs	SOAP, account credentials
S&S Activewear	Live pricing/inventory, catalog import, electronic POs	REST, API key
Gmail (Workspace)	Mailbox activity monitoring	Service account, domain-wide delegation, **gmail.readonly + metadata-only** — no compose/send
Missive	Correspondence sync; team's actual email client	Direct API token, paced ~1 req/s, GET-only today
Dropbox	Customer artwork discovery/preview	Managed OAuth connector, read-only, path-authorized
OpenAI	Ask AI, quote extraction, correspondence search	Managed proxy today; swap to own key = env change only
PrintNode	Auto-printing workorders	API key
ShipStation / EasyPost	Shipping rate quotes	API keys
Resend	Mint Bucks customer emails	Managed connector; test-mode redirect until go-live
Clerk	Staff sign-in	Managed tenant + server-side allowlist/roles
Chipply	Web-store results	No API — email/CSV parsing with mandatory review

Safety & Guardrail Principles

- **AI reads are sandboxed**: Ask AI runs SELECT-only validated SQL as a dedicated no-login DB role that can see only 18 whitelisted tables, inside a read-only transaction with timeout and row caps. Correspondence AI applies mailbox ACLs before retrieval and validates citations.
- **AI never writes to live systems**: quote drafts, suggestions, and enrichment always land in a review queue; a human types an explicit confirmation to execute.
- **Non-idempotent Printavo writes get one attempt**; ambiguous results lock for manual verification rather than retrying.
- **Kill switches and phased ownership** on every automation (auto-print, status rules, pollers, email mode) — everything ships OFF/dry-run and is flipped deliberately, one environment at a time.
- **Module isolation** is a hard requirement: each module is independently tunable without breaking others; Reports reads Postgres only, never the Printavo API.
- **Real-write testing uses designated test order #22251 exclusively — status changes are customer-visible in Printavo.**

Roadmap

NEXT — drafted, awaiting acceptance

- **Missive AI context layer (Task #44)** — the current focus and the reason for this review. Missive's native AI already triages and drafts replies well; the Hub will supply **verified Printavo knowledge** behind it via a Hub-hosted MCP server that Missive's assistant calls: confidence-scored customer matching (never a silent guess), read-only tools for orders/invoices/payments/mockups/artwork/brand standards/links, full tool-call audit logging, conversation-contact scoping. Verified constraints: Missive AI Rules cannot call MCP tools (only four fixed AI actions), so Pattern A ships first — rule drafts, staff runs a saved assistant command that enriches the draft with Hub facts; Pattern B (webhook → Hub posts a context note → rule drafts from it) is a timing spike. Phase 3 adds one safe action: create a Hub AI Quote Draft (enters the existing review/confirm flow — never a live Printavo quote). Explicitly excluded: sending email, live Printavo writes, exposing mailbox content through the shared MCP credential.
- **Complete customer Brand Standards (Task #45)** — copyable Pantone/HEX/RGB/CMYK + ink/thread values, structured typography, logo usage rules, approval history, printable standards sheet.

LOGGED — not scheduled

- **Shop knowledge base** (FAQ, SOPs, policies) structured for both humans and AI — the natural second context source for the Missive assistant; location (Hub-native vs Notion) undecided.
- **CFO AI assistant** — analyst agent over Printavo mirror + imported payroll/labor + email-effort data; tunable cost model; advisory digests on the Ask AI guardrails.
- **Sales & follow-up tools** — proactive follow-up reminders, receivables tracking, customer-activity alerts (get much smarter once correspondence data matures).
- **External analytics hub** — Fireflies, JustCall, Google Analytics, Meta, RB2B, etc. behind the existing role tiers.
- Offsite backup copies (Drive/NAS) · preset reports library · shared color catalog / key-authed catalog API for the sister BreezyTweak SaaS · outgoing-activity audit log · maintenance log · public reorder portal (last, after internal modules settle).

Prepared Aug 29, 2026 for external technical review of the Missive AI direction. Companion documents: Project Plan & Status sheet (module-by-module test status) · Missive/Printavo MCP architecture review (reviewed and folded into Task #44 with corrections: MCP endpoint must be served from the published deployment; the shared org-level MCP credential cannot honor per-staff mailbox ACLs, so tools expose Printavo/branding data only; existing customer-matching and quote-draft safety rails will be reused rather than rebuilt).